

Three archives, one instrument: the bookkeeping of Haghia Triada, Susa and Uruk measured by distribution

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Abstract

Three early administrative archives written in scripts whose languages are not understood, or understood only in part, are measured with a single set of distributional instruments: the Linear A tablets and roundels of Haghia Triada (Crete, c. 1450 BC; 1,117 sign-group tokens, 224 documents with two or more sign-groups), the Proto-Elamite tablets of Susa (Iran, c. 3100-2900 BC; 8,282 entries, 1,594 tablets) and the proto-cuneiform tablets of Uruk IV-III (Mesopotamia, c. 3350-3000 BC; 23,272 entries with numerals, 6,387 tablets). The same quantities are computed in each: the anatomy of the entry, the rate of singletons by length of sign-string, the behaviour that separates counted things from parties (recurrence within a document with different quantities), headings and totals, the exclusivity of vocabulary by document type, co-occurrence and co-exclusion against a fixed-margin null, the numeral systems by object class, and, where hands are attributed, the share of vocabulary explained by the scribe. The three archives share an anatomy (heading, entries of sign-string plus quantity, totals) and a partition of their sign-groups into a closed class of short, repeated commodity terms and an open class of long, singleton names; they differ in what each corpus can show at its size. Co-exclusion between document types, invisible at Haghia Triada, is measurable at Susa (13 pairs), and the inheritance of numeral systems from Uruk to Susa can be tested by structure (sexagesimal inherited, capacity adapted, decimal invented, bisexagesimal not taken). At Haghia Triada, uniquely, the attribution of hands allows the vocabulary to be assigned to scribes rather than to document types. The comparison places Minoan bookkeeping within the typology of early Near Eastern administration without reading a word, and gives, for each instrument, the corpus size at which it begins to see.

Keywords: Linear A, Proto-Elamite, proto-cuneiform, administrative archives, null models, co-exclusion, numeral systems, scribal hands.

1. Why compare three archives that cannot be read

The three corpora share the condition that makes them difficult and the one that makes them comparable. None has a bilingual: Linear A is not read; Proto-Elamite is not read and its language is unknown; proto-cuneiform is half read, through the continuity of its logograms and numerals into Sumerian cuneiform (Nissen, Damerow and Englund 1993; Englund 1998). All three are, on the other hand, accounting archives of a palace or institution, with the same document type: the tablet that records entries of a quantity of something associated with someone, with a heading and sometimes a total. The literature of each has described that anatomy separately (Schoep 2002 and Montecchi 2010 for Haghia Triada; Dahl 2005, 2019 and Damerow and Englund 1989 for Susa; Englund 1998 for Uruk), but it has not been measured

with the same instrument in all three, and so it is not known which features belong to early bookkeeping in general and which to each tradition.

The comparison also has a methodological value. The apparatus used here (Acedo 2026c) produces on Linear A a series of negative results that were attributed to the size of the corpus: no measurable co-exclusion, no reliability of network descriptors, no power to separate document types. Applied to sister corpora five and twenty times larger, it allows one to say for each instrument at what size it begins to see, and thus to turn those negatives into a power curve.

2. Data and instruments

Linear A: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer (Hogan 2022), with the hands of GORILA V and Younger's commentaries (2024); verification in SigLA (Salgarella and Castellan 2020). Proto-Elamite and proto-cuneiform: the transliterations and catalogue of CDLI (export of September 2026), with the numeral systems according to Damerow and Englund (1989) and Dahl (2019). In all three, the unit is the entry (a numbered line with a sign-string and a quantity); the sign-string of Linear A is the syllabic sign-group, that of Proto-Elamite and proto-cuneiform the sequence of non-numerical signs of the entry.

Instruments, in the same order in the three corpora: (1) size and singleton rate of the strings; (2) singletons by string length; (3) recurrence within a document with different quantities by length class; (4) headings (first entry) and totals; (5) vocabulary monopolies by document type or by numeral system; (6) co-occurrence and co-exclusion with the curveball null (Strona et al. 2014) on the documents $\times$ frequent strings matrix; (7) numeral systems by object class; (8) hands, where they exist. Nulls and thresholds are those described in Acedo (2026c): permutation preserving margins or lengths, $p < 0.01$ with one thousand permutations for new claims.

3. Results

3.1 The anatomy is the same

The three archives write the entry as sign-string plus quantity, open with a heading (SA-RA₂ and the toponyms at Haghia Triada; M157, the institution sign, on 278 tablets at Susa; the institution and office signs at Uruk) and close with a total on part of the tablets (KU-RO at Haghia Triada; the totals of Susa and Uruk marked by position). The proportion of entries with a quantity is of the same order (Haghia Triada 0.6; Susa 0.75; Uruk 0.8 of numbered lines).

3.2 The partition of the lexicon is the same, and is seen better with more data

In the three corpora short strings are few, frequent and repeated, and long strings are many, unique and not repeated. At Susa the gradation is clean: one-sign strings are singletons in 3% of cases and recur within the tablet with different quantities in 13%; strings of three or more signs are singletons in 87-99% and recur in 0.2%. At Haghia Triada the same partition exists (commodity and transaction terms recur, names are singletons at 76%) but is seen with less definition because the short class is mostly logographic. Uruk behaves like Susa. The partition between what is counted and who delivers or receives it is therefore a feature of early

bookkeeping and not of one tradition, and the behavioural criterion (recurrence with different quantities) recovers it without reading in all three.

3.3 The compartments, and who makes them

At Haghia Triada, the accounts with a total have six monopoly words and the allocations three ($p < 0.01$ in both cases); at Susa, the capacity system (grain) has 24 monopoly strings, the decimal 3, the sexagesimal none. Co-exclusion, which at Haghia Triada is not measurable (zero pairs with expected ≥ 1 in 224 documents), appears at Susa with 830 tablets: 13 pairs never co-occur when they should (the header M157 with M371, M370, M373, M046, M367; the grain product M297 with M376, M124, M003, M032, M009), which separates grain accounts from those of livestock and personnel. Only Haghia Triada allows the control by hand: with 73 tablets attributed to 21 scribes, shared vocabulary is explained by the hand (effect 0.032, $p = 0.002$ controlling for type) and not by the document type (0.025, $p = 0.21$): the Minoan compartments are, at least in part, scribal portfolios. Susa and Uruk have no attributed hands in the catalogue, and their compartmentation cannot be decomposed in this way; it is stated.

3.4 Long strings have a frame

At Susa, the 1,296 strings of three or more signs with a quantity have an institution or object sign in final position above chance (M288 231 times, lift 3.8, $p < 10^{-6}$; also M263, M297, M346) and a class sign in initial position (M124, M305, M157). The “names” carry a frame: classifier in front, variable core, institution or object behind. At Haghia Triada the frame exists with less power: the prefix KA- (17% of roundel groups against 4% of tablet groups) and the suffix -JA (seven root/derivative pairs against 3.1 expected). It is the same phenomenon at two scales of data.

3.5 The numerals

Susa inherits from Uruk the sexagesimal system (identical chain $N45 \rightarrow N34 \rightarrow N14 \rightarrow N01$, same carries), adapts the capacity system (same upper segment $N01 \rightarrow N39 \rightarrow N24$, its own fraction segment $N30C, N30D, N39C$), invents the decimal ($N23$ and the chain $N23 \rightarrow N14 \rightarrow N01$ do not exist at Uruk; $N51$ changes value, from 120 bisexagesimal to 1,000 decimal) and does not take the bisexagesimal (Uruk’s chain $N19 \rightarrow N04$, 231 cases, absent at Susa). It is the description of Damerow and Englund and of Dahl, recovered by structure, with the correction that the bisexagesimal inheritance is not confirmed. Linear A has no partner for comparison on this side: its fractions (Corazza et al. 2021) are a system of its own.

3.6 The periphery

At Susa the institutional header M157 is exclusive to the centre (278 tablets, 0 at Malyan, Yahya, Sialk or Sofalin); Yahya writes almost only grain accounts and has six signs of its own; Malyan uses the Susa inventory without a foreign sign. In Crete, Haghia Triada against Khania and Zakros shows a difference of repertoire of the same kind (the HT monopolies do not appear outside), with the same limitation of size of the periphery. In both cases the centre/periphery difference is one of administrative genre before one of script.

4. Discussion

What is common. The early bookkeeping of three regions without direct contact (Uruk and Susa had it; Crete with neither) has the same anatomy and the same partition of the lexicon, and in the three cases the part that records persons and places is the open class of long, unique strings, and the part that records what is counted is the closed class of short, repeated signs. That is not linguistics: it is the shape a record of deliveries takes when it is written, and it is seen in the three scripts because the three were born for that.

What is proper to each. Susa invents a decimal system for persons and animals and does not take the bisexagesimal; Crete writes the commodity almost always with a logogram and the party with syllables, and separates receipts (roundels) from lists (tablets) with a prefix; Uruk has the systems of time, rations and area that the other two do not need. Each archive shows its economy in what it decided to count and in how it decided to count it.

What it teaches about the instrument. Co-exclusion, the frame of names and the clean partition by length appear clearly at Susa and Uruk and not at Haghia Triada, whose size is a fifth and whose lexicon is mostly logographic. The control by hand, on the other hand, is possible only at Haghia Triada, because only there are scribes attributed. Each corpus answers the instruments that its size and its metadata allow, and the comparison fixes for the first time, with a number, the order of magnitude at which each instrument begins to see: co-exclusion, around eight hundred samples with mixed vocabulary; frame of names, around a thousand long strings; control by hand, with twenty hands and seventy documents.

What it does not say. Nothing about the languages, which remain three unknowns, nor about kinship between the scripts, which have none save Uruk-Susa. What it says is where each archive stands in the typology of early written administration, and that Linear A, the smallest of the three, is a normal archive of that typology, measured with the yardstick its elder siblings allow one to calibrate.

Data and reproducibility

Corpora, scripts and tables accompany the paper. Linear A: [lineara.xyz](#) (Hogan 2022) and [SigLA](#); Proto-Elamite and proto-cuneiform: [CDLI](#) ([cdli.earth](#)), exports of September 2026 included.

Declaration of assistance

Corpus processing, permutation tests and the first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

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